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Research Paper

Operation optimization of battery thermal management systems based on transient heat transfer model and self-adaptive control strategy

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ABSTRACT

Air-cooled battery thermal management systems (BTMSs) are critical to ensure the safety of battery packs in electric vehicles. Previous operation optimization of air-cooled BTMS relying on computational fluid dynamics (CFD) method for temperature evaluation is difficult to achieve the real-time regulation of temperature distribution of battery packs due to its long calculation time. In this work, an efficient transient heat transfer model is developed for operation optimization via flow pattern switch of BTMSs with different flow patterns. This prediction model outperforms the existing models, with the deviations within 7% between the obtained results and CFD results for different BTMSs. Based on this model, a control strategy is adopted to carry out the operation optimization in a BTMS consisting of J-type, U-type and L-type flow patterns, which allows the switch of the flow pattern in response to the predicted real-time temperature distribution. Even under high-rate discharge condition and varying operating conditions, the temperature difference of the battery pack can be controlled within 1 K for different conditions by combining the self-adaptive control strategy and the developed model, which is also validated by CFD method. The developed transient heat transfer model shows great potential for efficient performance evaluation and operation optimization of air-cooled BTMSs.

1. Introduction

Under the pressure of energy shortage and environmental pollution, electric vehicles are developed rapidly in recent years. Lithium-ion battery is widely used in electric vehicles as power sources, and its performance strongly depends on the operation temperature. The appropriate operating temperature for lithium-ion battery is within 0°C and 40°C, and the temperature difference inside battery pack should be within 5°C [1]. The large amount of heat generated during the charging or discharging process of batteries can easily overheat the battery and reduce its performance and service life. Therefore, it is necessary to design battery thermal management systems (BTMSs) so that battery packs can operate in a suitable temperature range.

The existing battery thermal management technologies include air cooling [2,3], liquid cooling [4–6], phase change material cooling [7–9], heat pipe cooling [10,11], and hybrid cooling [12,13]. Although the specific heat capacity of air is small, air cooling is still adopted in some EVs due to the advantages of simple structure and low cost, including Fenggon MINI EV, Nissan Leaf 2018, Volkswagen *E*-Golf 2017, Renault Zoe40 2017, Chevrolet Bolt EV, and Toyota Prius Prime [14]. However, large temperature difference will occur inside battery packs if the design of thermal management system is unreasonable. In order to improve the performance of air-cooled BTMS, some researches focus on the structural design of the systems. Saechan et al. [15] compared inline, offset and staggered arrangements of the battery cells, and found that the battery pack with staggered arrangement exhibited minimum temperature rise and temperature difference. Kirad et al. [16] found that increasing the longitudinal spacing of the cells can reduce the maximum temperature and temperature difference of the battery pack in the serial air-cooled system. Peng et al. [17] studied the battery pack with rectangular layout in a serial air-cooled system, and found that reducing the

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Nomenclature

C_μ, C_1 and C_2	parameters of k-ε turbulence model, 1
c_p	specific heat capacity, $J \cdot kg^{-1} \cdot K^{-1}$
h	convective heat transfer coefficient, $W \cdot m^{-2} \cdot K^{-1}$
I	current, A
k	turbulent kinetic energy, $m^2 \cdot s^{-2}$
l	length of channel, m
L	length of the system in x direction without inlet and outlet, m
m	number of repeated measurements, 1
N	number of batteries in x direction, 1
p	Reynolds-average pressure, Pa
ΔP	pressure drop of the system, Pa
Q	flow rate, $m^3 \cdot s^{-1}$
R	equivalent internal resistance, Ω
SOC	state of charge, 1
S	heat transfer area, m^2
T	temperature, K
ΔT	temperature difference of battery pack, K
ΔT_{left}	temperature difference between i-th battery and the airflow in the left channel, K
ΔT_{right}	temperature difference between i-th battery and the airflow in the right channel, K
ΔT_{lim}	temperature difference threshold, K
Δt	time step, s
u_i, u_j	Reynolds-average velocity components, $m \cdot s^{-1}$
U	average velocity, $m \cdot s^{-1}$
x_i, x_j	cartesian coordinates, m
x_{max}	horizontal coordinate of the center of the battery with the highest temperature, m

Greek symbols

ρ	density, $kg \cdot m^{-3}$
λ	thermal conductivity, $W \cdot m^{-1} \cdot K^{-1}$
μ	dynamic viscosity, $N \cdot s \cdot m^{-2}$
μ_t	turbulent dynamic viscosity, $N \cdot s \cdot m^{-2}$
ε	turbulent kinetic energy dissipation rate, $m^2 \cdot s^{-3}$
$\Delta\varepsilon$	temperature difference deviation, K
σ_T, σ_k, σ_ε,	turbulent Prandtl number for T, k and ε, 1
ϕ	heat generation rate, $W \cdot m^{-3}$

Subscripts

a	parameter of air
b	parameter of battery
in	parameter of inlet
max	maximum value
out	parameter of outlet
p	test point

Acronyms

BTMS	battery thermal management system
CFD	computational fluid dynamics
PC	parallel channel

rectangular aspect ratio could improve the cooling performance of the system. Zhao et al. [18] designed a battery pack with an isosceles trapezoid layout. The maximum temperature and temperature difference of the battery pack with this arrangement decreased by 0.9°C and 1.17°C respectively compared with the battery pack with a rectangular layout. Singh et al. [19] studied the influence of the location and number of inlet and outlet on the cooling performance of a parallel air-cooled system, and suggested that the outlet should be placed at the top of the system. Zhang et al. [20] added a secondary vent at the top of the parallel air-cooled system and baffles in the parallel channels. The designed system reduced the maximum temperature and temperature difference of the battery pack by 2.2 K and 92%, respectively. Liu et al. [21] divided the parallel channels of the Z-type, U-type and J-type air-cooled systems into four groups, and adopted genetic algorithm to optimize the width distributions of parallel channels. The results showed that the maximum temperature and temperature difference of the battery packs in the optimized systems were lower than those in the original systems. Li et al. [22] adopted multivariate genetic algorithm to optimize the inlet width, outlet width, parallel channel widths and air flow rate of U-type air-cooled system. After optimization, the temperature difference and temperature standard deviation of the battery pack and the volume of the system were reduced by 51.9%, 70% and 34.2%, respectively. Zhang et al. [23] developed the flow resistance network model and simplified heat transfer model to calculate the flow rate distribution and battery temperatures in parallel air-cooled BTMSs, and adopted a heuristic method to optimize the parallel channel widths of the systems with different flow patterns, significantly reducing the maximum temperature and temperature difference of the battery pack. Zhang et al. [24] deduced the optimization equations for turbulent heat convection, and obtained the optimal flow field of the air-cooled BTMSs for minimizing the hot spot temperature of the battery pack. Based on the optimal flow field, the flow resistance network model was adopted to optimize the parallel channel width, effectively reducing the hot spot temperature and temperature difference of the battery pack. Lyu et al. [25] optimized the angles of inlet and outlet deflectors in Z-type air-cooled system using the flow resistance network model, and further optimized the parallel channel widths using genetic algorithm, achieving a reduction of temperature difference inside battery pack by 80%.

The structural design can effectively improve the cooling performance of the air-cooled BTMS under the setting operating condition. However, the operating conditions of electric vehicles are complicated, and the heat generation rate varies with the operating conditions. The designed BTMSs under fixed conditions may be difficult to meet the requirement under other operating conditions. Therefore, some researches focus on the operation optimization for BTMSs. He et al. [26] switched the flow direction of a serial air-cooled BTMS according to the position of the highest temperature, which reduced the power consumption and improved the temperature uniformity of the batteries. Zhang et al. [27] found that the reciprocating air-cooled BTMS can reduce the temperature difference of the battery pack by 60% compared with the unidirectional one. Mahamud et al. [28] used reciprocating air flow in the serial air-cooled BTMS to reduce the maximum temperature of the battery pack and improve the uniformity of the temperature distribution. The effects of reciprocating cycle and air flow velocity on the system performance were investigated. Zhuang et al. [29] adjusted the battery cell spacing and added spoiler columns in a serial air-cooled BTMS to improve the cooling performance of the system. On the basis of this structure, the reciprocating air flow was introduced, and the suitable reciprocation period was determined after considering the power consumption and temperature uniformity. Wang et al. [30] proposed a control strategy for reciprocating air flow, which automatically switched the air flow direction when the maximum temperature difference of a single battery reached 4.9°C, thus realizing the automatic control of the temperature difference. Recently, Liu et al. [14] proposed a strategy to switch the flow patterns of the parallel air-cooled systems (Z-type, U-type and J-type flow patterns) according to the temperature distribution inside the battery pack. Numerical results indicate that the temperature difference of the battery pack was controlled within 1.33 K. Chen et al. [31] proposed a novel BTMS with J-type, U-type and L-type

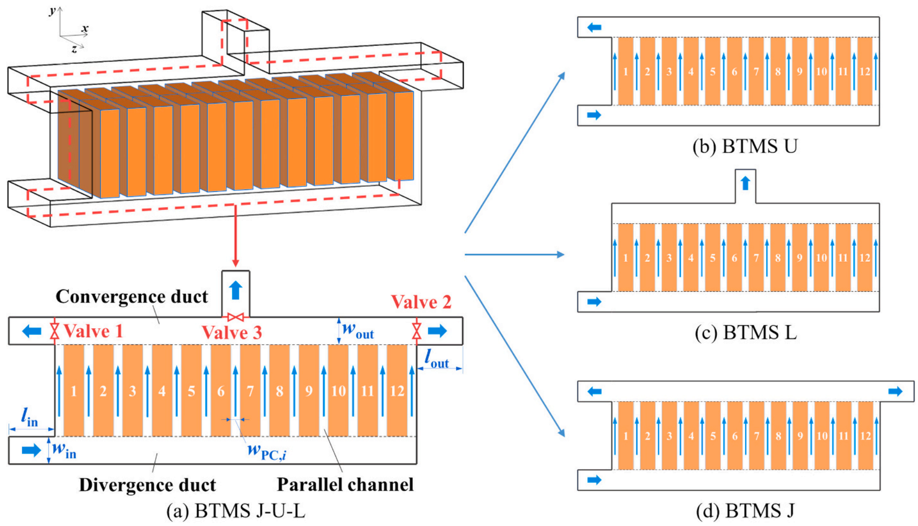


Fig. 1. Schematic of BTMS J-U-L.

flows, and developed a control strategy to switch the flow pattern according to the real-time temperature difference and the position of the hot spot temperature. The widths of the parallel channels were designed to improve the cooling performance of the system, and the numerical results showed that the temperature difference of the battery pack was always below the preset value after using the control strategy under varying operating conditions.

The previous studies demonstrated that the designed BTMSs combined with the control strategy can effectively improve the temperature uniformity of battery packs under different operating conditions. In these strategies, temperature prediction is achieved by using computational fluid dynamics (CFD) method to calculate the temperature field of battery packs, which is time-consuming due to the meshed calculation domain and the calculation process of solving the partial differential equations of velocity and temperature. Thus, the operation optimization from control strategy using CFD method is difficult to achieve the real-time regulation of the temperature difference inside the battery pack, resulting in the delay in operation. On the other hand, existing transient heat transfer models for air-cooled BTMSs still show a substantial error ($>$ 0.5 K), with poor accuracy in estimation of the temperature distribution of battery pack. Therefore, it is still challenging to develop an efficient transient heat transfer model for operation optimization of air-cooled BTMSs and realize the real-time flow pattern regulation of the system.

To address this issue, an efficient transient heat transfer model is developed for operation optimization via flow pattern switch of BTMSs with different flow patterns based on law of energy conservation. The effectiveness of the model is verified using CFD method. Based on the developed model, a self-adaptive control strategy is proposed to carry out operation optimization (flow pattern switch) for the typical BTMS to regulate the temperature difference inside the battery pack. The performance of the BTMS with the model and control strategy is evaluated under high-rate discharge condition and varying operating conditions. Finally, the operation optimization results are verified using CFD method, and are compared with the results of the system without flow pattern switch.

2. Typical BTMS with self-adaptive control strategy

2.1. Physical model of the BTMS

Previous study [31] has proposed the BTMS consisting of J-type, U-type and L-type (denoted as BTMS J-U-L) to cool battery packs. In this work, BTMS J-U-L is considered, as shown in Fig. 1a. This system contains one inlet and three outlets, and each outlet is controlled with one valve (denoted as Valve1, Valve2 and Valve3). By opening or closing the valves, different flow patterns can be switched for the system. When Valve1 is opened and both Valve2 and Valve3 are closed, the flow pattern is U-type flow (Fig. 1b). When Valve3 is opened and both Valve1 and Valve2 are closed, the flow pattern is L-type flow (Fig. 1c). When both Valve1 and Valve2 are opened and Valve3 is closed, the flow pattern is J-type flow (Fig. 1d). These three systems are denoted as BTMS J, BTMS U and BTMS L, respectively. In this study, the battery cell with size of 16 mm $\times$ 151 mm $\times$ 65 mm is considered, with the physical properties [32] shown in Table 1. There are 12 $\times$ 2 battery cells in the system. The inlet and outlet widths of the system are both 20 mm, and the lengths are both 100 mm. The widths of the parallel channels are [2.5, 3.1, 3.1, 3.1, 3.1, 3.1, 3.1, 3.1, 3.1, 2.5] (mm) [31]. The inlet flow rate Q_{in} is set to 0.015 $m^3 \cdot s^{-1}$, with the corresponding inlet Reynolds number at 12527. The inlet air temperature T_{in} and the initial system temperature T_0 are both 298.15 K.

Table 1
Properties of battery, aluminum block and air.

Property	Air	Battery [32]	Aluminum block
Density ($kg \cdot m^{-3}$)	1.165	1542.9	2700
Specific heat capacity ($J \cdot kg^{-1} \cdot K^{-1}$)	1005	1337	900
Dynamic viscosity ($kg \cdot m^{-1} \cdot s^{-1}$)	1.86×10^{-5}	–	–
Thermal conductivity ($W \cdot m^{-1} \cdot K^{-1}$)	0.0267	1.05(x),21.1 (y),21.1(z)	240

Considering the five-current (5C) discharge process for the battery cells, the heat generation rate of the battery is calculated using the model by experiments in the previous study [33], shown as:

$$\phi_b = \frac{1}{V_b}\left(I^2R - IT_b\frac{dU_e}{dT}\right) \tag{1}$$

$$\begin{gathered} R = 0.00705 - 0.01853 \times SOC + 0.05894 \times SOC^2 \\ -0.09151 \times SOC^3 + 0.06579 \times SOC^4 - 0.01707 \times SOC^5 \end{gathered} \tag{2}$$

where the subscript “b” represents the battery cell. I and R are the current and equivalent resistance of the battery, respectively. ϕ, V and T are the heat generation rate, volume and temperature. The coefficient dU_e/dT is set to -0.22 $mV \cdot K^{-1}$ [33].

Different parameters are adopted to evaluate the performance of the system, including the maximum temperature ($T_{\max}$), temperature difference of the battery pack (ΔT), and the power consumption of the system ($W_{\rm p}$). ΔT and $W_{\rm p}$ are calculated by the following expressions:

$$\Delta T = \max\{T_{\rm b,1}, \cdots, T_{{\rm b},i}, \cdots, T_{{\rm b},N}\} - \min\{T_{\rm b,1}, \cdots, T_{{\rm b},i}, \cdots, T_{{\rm b},N}\} \quad (3)$$

$$W_{\rm p} = Q_{\rm in}\cdot\Delta P \quad (4)$$

where $T_{{\rm b},i}$ represents the average temperature of the i-th battery cell, ΔP represents the pressure drop of the system.

2.2. Self-adaptive control strategy

In the previous study [31], a self-adaptive control strategy was developed to switch the flow pattern of the BTMS for regulating the temperature difference of the battery pack. This control strategy is introduced to regulate the temperature difference of the battery pack in BTMS J-U-L under different operating conditions. The idea of this control strategy is to switch the flow pattern for driving more cooling air to the parallel channels around the battery cells with high temperature. The detailed steps of the self-adaptive strategy are shown as follows.

The temperature difference threshold $\Delta T_{\lim}$, temperature difference deviation $\Delta\varepsilon$ and time step Δt are set. When the temperature difference of the battery pack $\Delta T > \Delta T_{\lim}$ - $\Delta\varepsilon$, the position of the battery with the highest temperature is found and the horizontal coordinate of the center of this battery is denoted as $x_{\max}$. The range of $x_{\max}$ is between 0 and L, where L is the length of the system in x direction (excluding the inlet and outlet segments). The following operations are adopted to control the valves at the outlets:

When $x_{\max} \le L/4$, Valve1 is opened, with Valve2 and Valve3 closed. The flow pattern is switched to U-type flow (Fig. 1b).

When $L/4 < x_{\max} \le 3L/4$, Valve3 is opened, with Valve1 and Valve2 closed. The flow pattern is switched to L-type flow (Fig. 1c);

When $3L/4 < x_{\max} \le L$, Valve1 and Valve2 are opened, with Valve3 closed. The flow pattern is switched to J-type flow (Fig. 1d).

By switching the flow pattern using the above strategy, more cooling air flows to the batteries with high temperature. Therefore, the hot spot temperature and temperature difference of the battery pack can be reduced. $\Delta\varepsilon$ is a deviation that is greater than the temperature difference rising in one time step, which can ensure that ΔT is below $\Delta T_{\lim}$ when switching flow patterns. In practical applications, the valves are controlled by control units. The reaction time of the control units will influence the delay time of flow pattern switching of the system. This impact can be eliminated through setting the value of temperature difference deviation $\Delta\varepsilon$. For example, if the delay time is $t_{\rm d}$, $\Delta\varepsilon$ can be set to a deviation that is greater than the temperature difference rising in ($t_{\rm d}$ + 1) time steps. The idea of this self-adaptive strategy is that when the temperature difference of the battery pack exceeds the preset value, the outlet nearest to the battery with maximum temperature is found, and more cooling air flows to high-temperature battery by switching the flow pattern, thus reducing the maximum temperature and temperature difference of the battery pack.

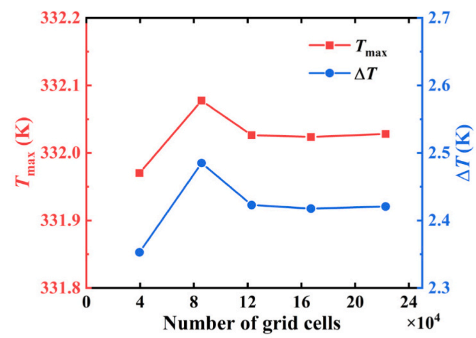
2.3. Numerical method

To evaluate the performance of the BTMS, Computational Fluid Dynamics (CFD) method is adopted to solve the flow and temperature equations. The air flow in the system is turbulent, so incompressible Navier-Stokes equations and standard k-ε model are introduced to calculate the flow field of the system. The governing equations are shown as follows:

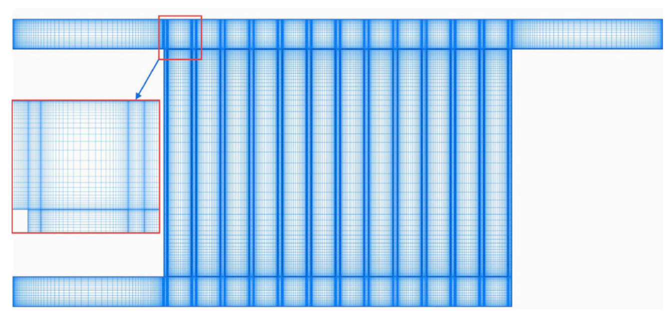
Continuity equation:

$$\frac{\partial u_j}{\partial x_j} = 0 \quad (5)$$

Momentum equation:



(a) Results of grid dependence analysis



(b) Grid system

Fig. 2. Results of grid dependence analysis and grid system of BTMS J.

$$\rho_{\rm a}u_j\frac{\partial u_i}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[(\mu_{\rm a}+\mu_{\rm t})\frac{\partial u_i}{\partial x_j}\right] \quad (6)$$

Turbulent kinetic energy equation:

$$\rho_{\rm a}u_j\frac{\partial k}{\partial x_j} = -\frac{\partial}{\partial x_j}\left[\left(\mu_{\rm a}+\frac{\mu_{\rm t}}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right] + \frac{\mu_{\rm t}}{2}\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right)^2 - \rho_{\rm a}\varepsilon \quad (7)$$

Turbulent kinetic energy dissipation rate equation:

$$\rho_{\rm a}u_j\frac{\partial \varepsilon}{\partial x_j} = -\frac{\partial}{\partial x_j}\left[\left(\mu_{\rm a}+\frac{\mu_{\rm t}}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_j}\right] + C_1\frac{\mu_{\rm t}}{2}\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right)^2 - C_2\rho_{\rm a}\frac{\varepsilon^2}{k} \quad (8)$$

$$\mu_{\rm t} = \rho_{\rm a}C_\mu\frac{k^2}{\varepsilon} \quad (9)$$

As the heat generation rate of the battery varies with time, unsteady energy equations are introduced to calculate the temperature field of the system:

Area region:

$$\rho_{\rm a}c_{\rm p,a}\frac{\partial T_{\rm a}}{\partial t} + \rho c_{\rm p,a}u_j\frac{\partial T_{\rm a}}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\lambda_{\rm a}+\frac{c_{\rm p,a}\mu_{\rm t}}{\sigma_T}\right)\frac{\partial T_{\rm a}}{\partial x_j}\right] \quad (10)$$

Battery region:

$$\rho_{\rm b}c_{\rm p,b}\frac{\partial T_{\rm b}}{\partial t} = \frac{\partial}{\partial x_j}\left(\lambda_{\rm b}\frac{\partial T_{\rm b}}{\partial x_j}\right) + \phi_{\rm b} \quad (11)$$

where the subscript "a" represents the cooling air. u_i and u_j are Reynolds-average velocity components, p is Reynolds-average pressure. ρ, $c_{\rm p}$ and λ are the density, specific heat capacity and thermal conductivity. $\mu_{\rm a}$ and $\mu_{\rm t}$ are molecular dynamic viscosity coefficients and turbulent dynamic viscosity coefficients, respectively. k and ε are the turbulent kinetic energy and turbulent kinetic energy dissipation rate respectively. C_1, C_2, C_μ, σ_k, σ_ε and σ_T are the parameters of the k-ε turbulence model. The values of the parameters are shown as follows [19]:

$$C_1 = 1.44, C_2 = 1.92, C_\mu = 0.09, \sigma_k = 1.0, \sigma_\varepsilon = 1.3, \sigma_T = 0.85 \quad (12)$$

In order to solve the governing equations, appropriate boundary

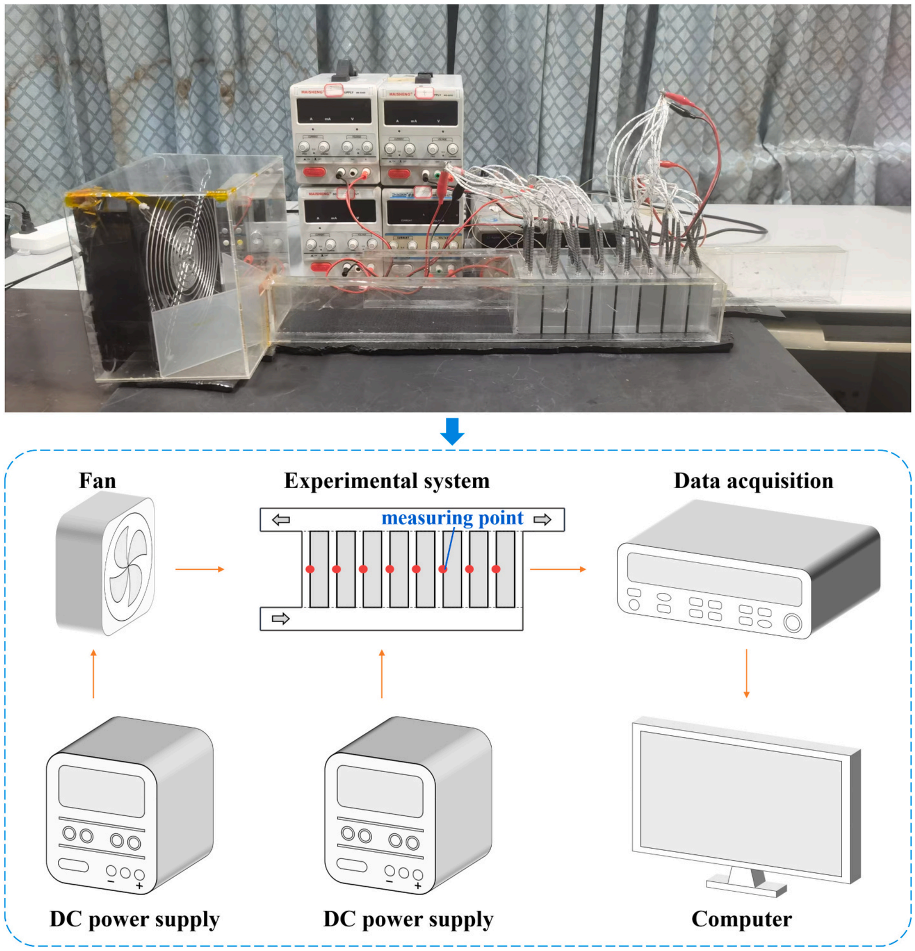


Fig. 3. Experimental system.

conditions are set. The inlet is set as the velocity inlet with constant temperature, while the outlet is set as the pressure outlet. The outer walls are set to no-slip walls with adiabatic condition, and the fluid-solid interfaces are set to no-slip coupled walls. The calculation domain is the same for the systems with different flow patterns. The flow pattern of the system is switched by changing the boundary conditions of the valves. When the valve is open, the boundary condition of the relevant valve is set to internal condition. When the valve is closed, the boundary condition of the relevant valve is set to no-slip wall. The governing equations are discretized using finite volume method, and solved using SIMPLE algorithm. The convergence residuals of the equations are set to 10^{-10}. The commercial software FLUENT is adopted to carry out the CFD calculation.

For the BTMSs investigated, the previous study [34] has shown that the numerical results of the two-dimensional system were consistent with those of the three-dimensional system. To save calculation time, two-dimensional simulation is adopted to evaluate the performance of the BTMS. To determine appropriate grid size, grid dependence test was performed. Taking BTMS J as an example, various grid numbers were used to discretize the computational domain. The simulation results are shown in Fig. 2a. It can be seen that when the number of grids exceeds 1.2×10^5, both the variations of ΔT and T_{max} are less than 0.01 K. Thus, the grid system with 1.2×10^5 cells is selected for the following CFD calculation, as shown in Fig. 2b.

2.4. Experimental validation of the CFD calculation

To verify the results of the CFD method, an experimental system with J-type flow pattern was established. Fig. 3 depicts the experimental system. Aluminum blocks were adopted to simulate the battery cells. There were eight blocks in the experimental system. K-type thermocouples were attached to the centers of the aluminum block's surfaces using copper-foil tape for measuring the local temperature. As the size of the K-type thermocouples is very small (< 1 mm in diameter), the measured temperatures can represent the surface temperatures of the blocks. Then the measured temperatures were recorded using the data acquisition instrument. The size of the aluminum blocks is 27 mm × 90 mm × 70 mm. The widths of the inlet and outlet ducts are both 20 mm, and the widths of the parallel channels were identical at 3 mm. Table 1 shows the properties of the aluminum blocks. Heat power of 12 W was applied to each aluminum block by three heating rods, and the heating rods were powered by the power supply. The inlet velocity of the system (U_{in}) was set to 2.5 $m \cdot s^{-1}$, which was measured using a hotwire anemometer. Table 2 lists the parameters of the applied instruments. The experiments were performed three times, and the uncertainty analysis was carried out, obtaining the uncertainties of the maximum temperature ($T_{max,p}$) and temperature difference (ΔT_p) of the test points as 0.9 K and 1.0 K respectively.

Table 2
Specifications of the experimental instruments.

Instrument	Specifications	Measuring range	Accuracy
K-type thermocouple	ETA TT-K-30	−200–260 °C	±1.1 °C
Data acquisition	Agilent 34970 A	–	±1%
Dc power supply	KXN-645D	Current: 0–5 A Voltage: 0–64 V	Current: ±1% Voltage: ±1%
Anemometer	KANOMAX 6004	0.1–20.0 m/s	±(5% + 0.1) m/s

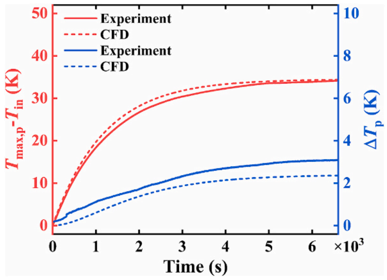


Fig. 4. $T_{max,p}$ and ΔT_p of the CFD method and experiment ($U_{in} = 2.5$ m/s).

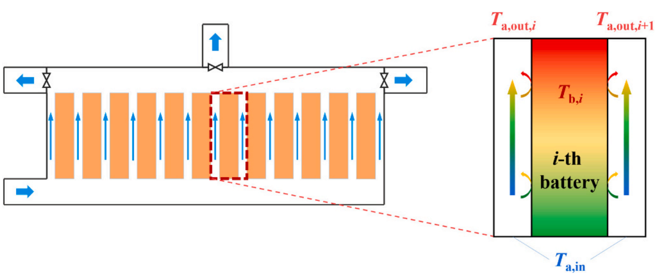


Fig. 5. Schematic of the convective heat transfer between a battery cell and air.

The CFD calculation was carried out for the experimental system with U_{in} as 2.5 m·s^{-1}. Fig. 4 shows the $T_{max,p}$ and ΔT_p values by the CFD calculation and experiment respectively. The results show that the average deviations of $T_{max,p}$ and ΔT_p between the CFD method and experiment are 0.9 K and 0.5 K, respectively. At the end of the heating process, the deviations of $T_{max,p}$ and ΔT_p between the two methods are 0.3 K and 0.8 K, respectively. In summary, the CFD method is verified by experiments, and can be adopted to predict the performance of the air-cooled BTMSs.

3. Transient heat transfer model

3.1. Equations of transient heat transfer model

Although the CFD method can accurately calculate the temperature field of the battery pack, it is time-consuming. Thus, it is difficult to realize the real-time prediction of the temperature difference inside the battery pack for the operation optimization of the BTMS. In order to improve the computational efficiency, a transient heat transfer model is developed to quickly calculate the real-time temperature distribution of the battery cells.

The convective heat transfer between the battery and the cooling air is shown in Fig. 5. Previous study [23] has developed a transient heat transfer model to calculate the real-time temperature of the battery cell. Based on the equations of CFD methods, the following assumptions are considered: (1) The temperature of each battery cell is uniform; (2) The air temperature in parallel channels satisfies linear distribution along the flow direction; (3) The inlet temperatures of the air are assumed identical for all parallel channels, which is equal to the inlet temperature of the system (T_{in}); (4) The heat transfer areas between the bottom/top of the batteries and the air in the divergence/convergence duct are equivalent to the area in the parallel channels. Therefore, for the *i*-th battery cell and the air in the *i*-th parallel channel, according to the law of energy conservation, the following equations are obtained [23]:

For the *i*-th battery cell

$$\rho_b c_{p,b} V_b \frac{\partial T_{b,i}}{\partial t} = \phi_{b,i} V_b - h_{PC,i} S \Delta T_{left,i} - h_{PC,i+1} S \Delta T_{right,i} \tag{13}$$

For air in the *i*-th parallel channel

$$\frac{1}{2}\rho_a c_{p,a} V_{PC,i} \frac{\partial T_{a,out,i}}{\partial t} + \rho_a c_{p,a} Q_{PC,i}\left(T_{a,out,i} - T_{a,in}\right) = h_{PC,i} S \Delta T_{left,i} + h_{PC,i} S \Delta T_{right,i-1} \tag{14}$$

where S is the equivalent heat transfer area between the battery and the air, including the areas of the parallel cooling channel, divergence duct and convergence duct, and h is the heat transfer coefficient. $T_{a,in}$ and $T_{a,out,i}$ are the air temperatures at the inlet and outlet of the *i*-th parallel channel respectively. $\Delta T_{left,i}$ is the temperature difference between the *i*-th battery cell and the airflow in the left channel. $\Delta T_{right,i}$ is the temperature difference between the *i*-th battery and the airflow in the right channel. These two temperature differences are calculated by

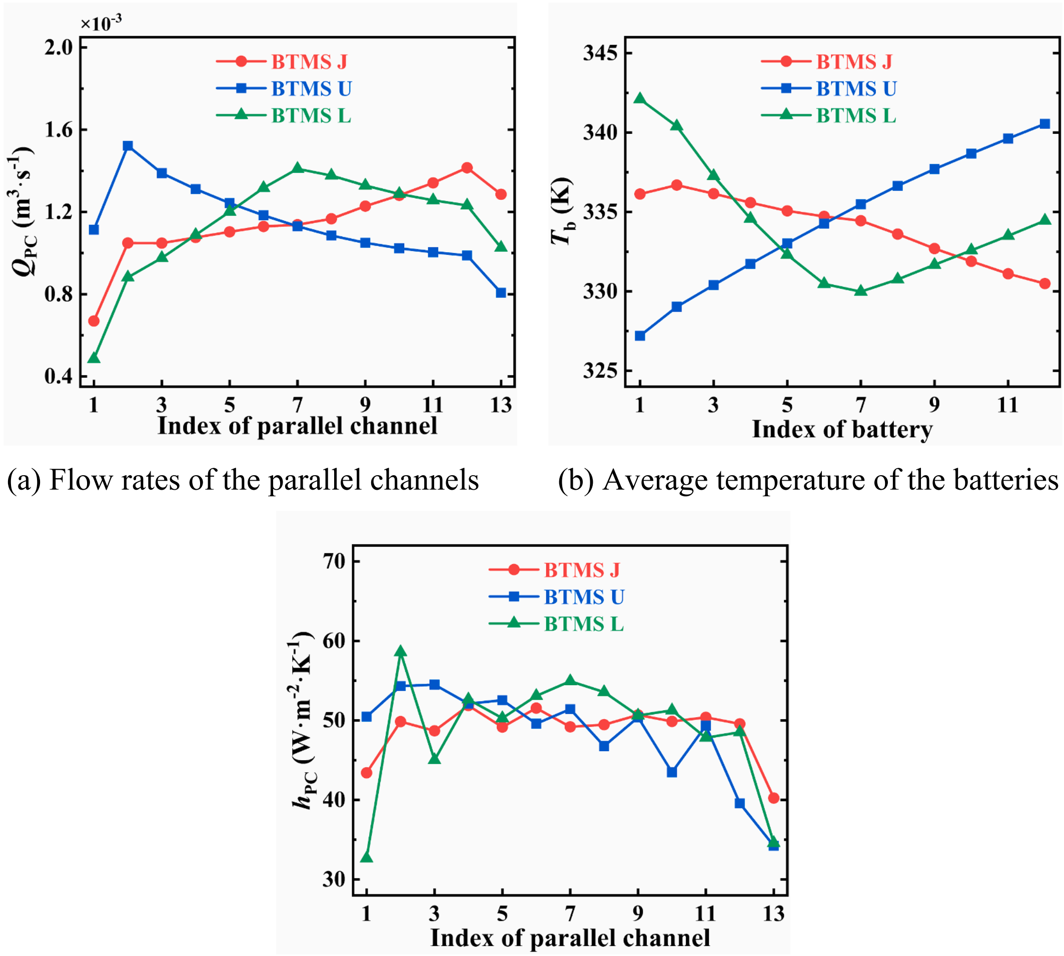
$$\Delta T_{left,i} = \begin{cases} \dfrac{\left(T_{b,i} - T_{a,in}\right) - \left(T_{b,i} - T_{a,out,i}\right)}{\ln\left(T_{b,i} - T_{a,in}\right) - \ln\left(T_{b,i} - T_{a,out,i}\right)}, & T_{b,i} > T_{a,out,i} \\ \dfrac{0.5^{\frac{1}{0.3275}}}{T_{a,out,i} - T_0}\left[\left(T_{b,i} - T_{a,in}\right)^2 - \left(T_{a,out,i} - T_{b,i}\right)^2\right], & T_{b,i} \le T_{a,out,i} \end{cases} \tag{15}$$

$$\Delta T_{right,i} = \begin{cases} \dfrac{\left(T_{b,i} - T_{a,in}\right) - \left(T_{b,i} - T_{a,out,i+1}\right)}{\ln\left(T_{b,i} - T_{a,in}\right) - \ln\left(T_{b,i} - T_{a,out,i+1}\right)}, & T_{b,i} > T_{a,out,i+1} \\ \dfrac{0.5^{\frac{1}{0.3275}}}{T_{a,out,i+1} - T_{a,in}}\left[\left(T_{b,i} - T_{a,in}\right)^2 - \left(T_{a,out,i+1} - T_{b,i}\right)^2\right], & T_{b,i} \le T_{a,out,i+1} \end{cases} \tag{16}$$

The initial temperature boundary conditions of the transient heat transfer model are the same as those of the CFD method. The boundary conditions are equivalent to the heat transfer term in Eq. (14). For the adiabatic boundary condition of the left border of the first parallel channel, $\Delta T_{right,0} = 0$. For the adiabatic boundary condition of the right border of the $(N+1)$ parallel channel, $\Delta T_{left,N+1} = 0$. In the above heat transfer model, for a system with N battery cells, the unknown variables to be solved are $T_{b,i}$ ($i = 1, 2, \cdot\cdot, N$) and $T_{a,out,i}$ ($i = 1, 2, \cdot\cdot, N+1$). The total number of the unknown variables is $2N+1$. The model provides $2N+1$ equations. Therefore, when the convective heat transfer coefficients $h_{PC,i}$, flow rates in parallel channels $Q_{PC,i}$ and the inlet temperature $T_{a,in}$ are given, the battery temperatures $T_{b,i}$ can be calculated using the transient heat transfer model. In the existing model, the convective heat transfer coefficients $h_{PC,i}$ are calculated using correction expressions of Nusselt number. However, there exist errors for the correction expressions, which leads to the errors in the heat transfer model. Therefore, it needs to obtain more accurate $h_{PC,i}$ to improve the accuracy of the model.

3.2. Calculation of convective heat transfer coefficient

In this section, an effective method for calculating convective heat transfer coefficients in the transient heat transfer model is developed based on the CFD results. According to the aforementioned analysis, when the convective heat transfer coefficient $h_{PC,i}$, parallel channel flow rate $Q_{PC,i}$ and air inlet temperature $T_{a,in}$ are given, the battery temperature $T_{b,i}$ can be calculated using the transient heat transfer model. Conversely, if the battery temperature $T_{b,i}$, parallel channel flow rate $Q_{PC,i}$ and air inlet temperature $T_{a,in}$ are given, the convective heat transfer coefficients $h_{PC,i}$ can be calculated using the model. In this study, the properties of air are considered as constants due to the small temperature difference of the air in the system, thus the convective heat transfer coefficients $h_{PC,i}$ are considered as constants. When the inlet flow rate and the structure of the system are fixed, $Q_{PC,i}$ can be calculated using CFD method. Assuming that the heat generation rate of the battery is a constant, the battery temperature $T_{b,i}$ in BTMSs can be also calculated using CFD method. When calculating $h_{PC,i}$, for a system with N battery cells, the number of the variables to be solved ($h_{PC,i}$ and $T_{a,out,i}$) is $2(N+1)$, and the number of the equations for the model is only $(2N+$



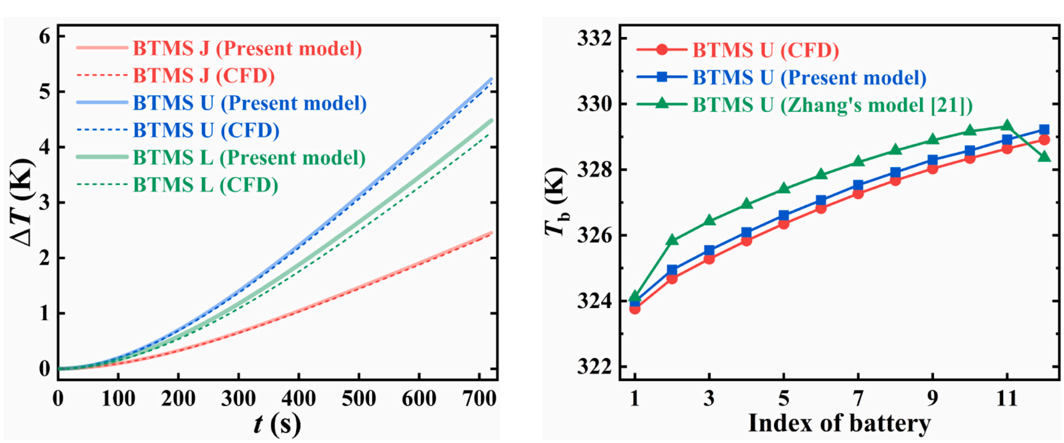
(a) Flow rates of the parallel channels (b) Average temperature of the batteries

(c) Convective heat transfer coefficient in parallel channels

Fig. 6. Comparison of BTMSs with different flow patterns.

1). The number of the unknown variables is one more than the number of the equations. It needs to add one equation to make the calculation closed. In this study, the convective heat transfer coefficient in one parallel channel is calculated using CFD method, and $h_{PC,i}$ in other parallel flow channels are treated as the variables to be solved, which can be calculated using the transient heat transfer model. To reduce the calculation error, the parallel channel with the smallest flow rate is selected and its $h_{PC,i}$ value is calculated.

CFD calculations are carried out for BTMS J, BTMS U and BTMS L, respectively. Fig. 6a shows the air flow rate in each parallel channel ($Q_{PC,i}$), and Fig. 6b shows the average temperature of each battery cell ($T_{b,i}$). It can be observed from Fig. 6a that the maximum flow rate is in



(a) Variation of ΔT with time (b) Battery temperature at t =720 s

Fig. 7. Comparison of the transient heat transfer model results and CFD results.

Table 3
Comparison of the heat transfer model results and CFD results.

	ΔT_{CFD}(K)	ΔT_{model}(K)	$E_{\Delta T}$(K)	$\eta_{\Delta T}$	E_T(K)
BTMS J	2.4	2.5	0.1	4%	0.3
BTMS U	5.1	5.2	0.1	2%	0.3
BTMS L	4.3	4.6	0.3	7%	0.1

the right channel for BTMS J, is in the left channel for BTMS U, and is in the middle channel for BTMS L. Therefore, by using different flow patterns in BTMS J-U-L, more cooling air can be moved to the channel at different position. Moreover, the minimum flow rate is in the 1st parallel channel for BTMS J and BTMS L, and is in the $N+1$ parallel channel for BTMS U. Therefore, for BTMS J and BTMS L, the convective heat transfer coefficient in the 1st channel is solved using CFD method, while the one in the $N+1$ channel is solved for BTMS U. Using CFD method, the average flow rate $Q_{PC,i}$, the outlet temperature ($T_{a,out,i}$) of the selected channel, and the average temperature $T_{b,i}$ of the battery cells adjacent to this channel can be obtained. By substituting these results into Eq. (14), the convective heat transfer coefficients of the selected channel can be obtained, which are 43.4 $W\cdot m^{-2}\cdot K^{-1}$, 34.2 $W\cdot m^{-2}\cdot K^{-1}$ and 32.6 $W\cdot m^{-2}\cdot K^{-1}$, respectively.

When the convective heat transfer coefficient of the selected channel is obtained, other $h_{PC,i}$ and $T_{a,out,i}$ can be calculated using the heat transfer model. Fig. 6c shows the calculated convective heat transfer coefficients in parallel channels for the three BTMSs. The obtained $h_{PC,i}$ are then substituted into the heat transfer model, and the model can calculate the average temperatures of the battery cells for the three BTMSs. Moreover, it is noticed that the $h_{PC,i}$ values depend on inlet flow rates and the structural parameters of the system. If these parameters are changed, the $h_{PC,i}$ values should be recalculated using the developed method in this section.

3.3. Validation of the model

In order to verify the accuracy of the transient heat transfer model, the model results are compared with the CFD results. Fig. 7a shows the temperature differences inside the battery pack (ΔT) of different BTMSs by both the heat transfer model and CFD method. It can be observed that the model results are in good consistency with the CFD results for different BTMSs. The deviations of the model results increase with time. The maximum deviation occurs at the end of the discharge process (t = 720 s). Table 3 lists the ΔT values obtained from the model and CFD method when the discharge process is finished. For the three systems, the absolute deviations ($E_{\Delta T}$) are 0.1 K, 0.1 K and 0.3 K, respectively, and the relative deviations ($\eta_{\Delta T}$) are only 4%, 2% and 7%, respectively. Fig. 7b shows the battery temperatures ($T_{b,i}$) when the discharge process is finished. It can be seen that the $T_{b,i}$ values by the developed model agree well with the CFD results. The average deviations of $T_{b,i}$ (E_T) are 0.3 K, 0.3 K and 0.1 K for the three BTMSs. The $T_{b,i}$ results of BTMS U from the previous transient heat transfer model by Zhang et al. [23] are also listed in Fig. 7b. The results indicate that the developed transient model is more accurate than the previous one. The average deviation of the battery temperatures (E_T) using the developed model decreases by 67% compared to those using the previous model. Furthermore, the model needs much less calculation time than the CFD method. The proposed transient model needs 1 s to obtain the results of 720 s, while the full CFD simulations need 20 min to obtain the results. Therefore, the developed transient heat transfer model can be applied for evaluating the real-time temperature difference inside the battery pack for BTMSs with different flow patterns.

3.4. Improved strategy based on the model

According to the results in Section 3.3, there is a deviation between the model results and the CFD results. Considering the deviation of the

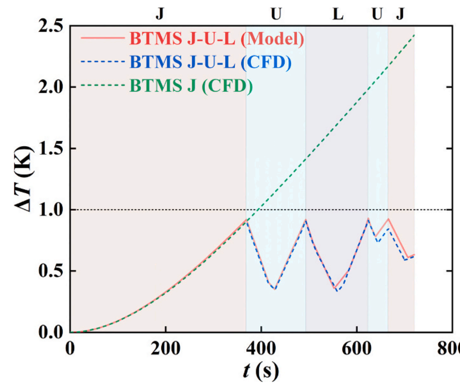


Fig. 8. Variation of ΔT with time under 5C discharge condition ($\Delta T_{lim} = 1.0$ K).

transient heat transfer model, a parameter δ is adopted to modify the criteria of flow pattern switching, which ensures that the temperature difference inside the battery pack is below the preset value during the discharge process. The modified criterion is shown as.

$$\Delta T > \Delta T_{lim}(1-\delta) - \Delta\varepsilon \tag{17}$$

That is, when $\Delta T > \Delta T_{lim}(1\text{-}\delta)$ - $\Delta\varepsilon$, the flow pattern is switched according to the self-adaptive control strategy in Section 2.2. As can be seen from Table 3, the maximum relative deviation of ΔT is no more than 7% for the three flow patterns in BTMS J-U-L. Therefore, δ is set to 7%.

4. Results and discussion

4.1. Condition of 5C high discharge rate

Considering the discharge process with a constant discharge rate of 5C, the discharge time is 720 s. The heat generation rate of the batteries is calculated by Eq. (1) in Section 2.1. The transient heat transfer model is used to calculate the average temperature of each battery cell and the temperature difference (ΔT). Combined the developed heat transfer model with the modified self-adaptive control strategy, the flow pattern of the system is switched according to the real-time temperature distribution within the battery pack, and is expected to control ΔT within the preset value during the whole operation process.

ΔT_{lim}, $\Delta\varepsilon$, Δt and δ are set to 1 K, 0.01 K, 1 s and 7%, respectively. The control strategy combined with the developed transient heat transfer model is adopted to switch the flow pattern of BTMS J-U-L. The CFD results show that the ΔT values of BTMSs J, U, and L are 2.4 K, 5.1 K and 4.3 K respectively, and the ΔP values are 20.4 Pa, 41.0 Pa and 48.3 Pa respectively. The J-type system achieves the lowest ΔT and ΔP among the three systems, so the initial flow pattern is set to J-type. Fig. 8 shows the variations of the ΔT value with time under the control strategy. It can be seen that ΔT reaches the criterion of flow pattern switch ($\Delta T_{lim}(1\text{-}\delta)$ - $\Delta\varepsilon$ = 0.92 K) at t = 369 s, and the battery with the maximum temperature is battery #2. Thus, the flow pattern is switched from J-type flow to U-type flow. After the switch, ΔT decreases remarkably first, and then gradually increases again. When t = 493 s, ΔT reaches the criterion value 0.92 K again, and the flow pattern is switched to L-type. Again, ΔT decreases significantly, and then increases. At t = 624 s, ΔT reaches 0.92 K again, and the flow pattern is switched to U-type flow. Finally, at t = 666 s, ΔT reaches the criterion of flow pattern switch, and the flow pattern is switched to J-type flow. After this switch, ΔT is below 0.92 K until the discharge process finishes (720 s). By using the self-adaptive control strategy and the transient heat transfer model, ΔT remains below 1.0 K during the whole discharge process after switching the flow pattern four times.

Then CFD calculation is carried out according to switch time and flow patterns of the control results, and the ΔT values with time are also

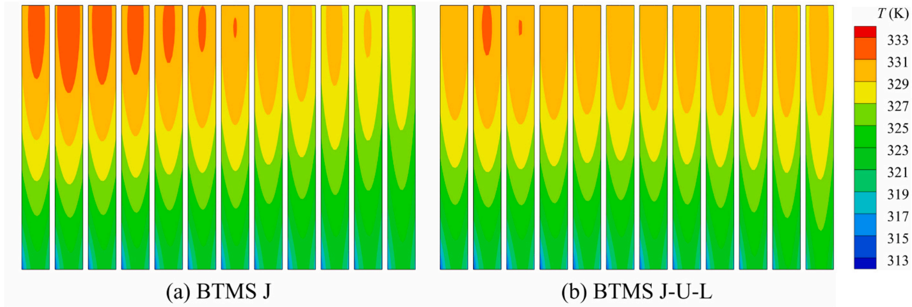


Fig. 9. Temperature contours of the battery pack under 5C discharge condition.

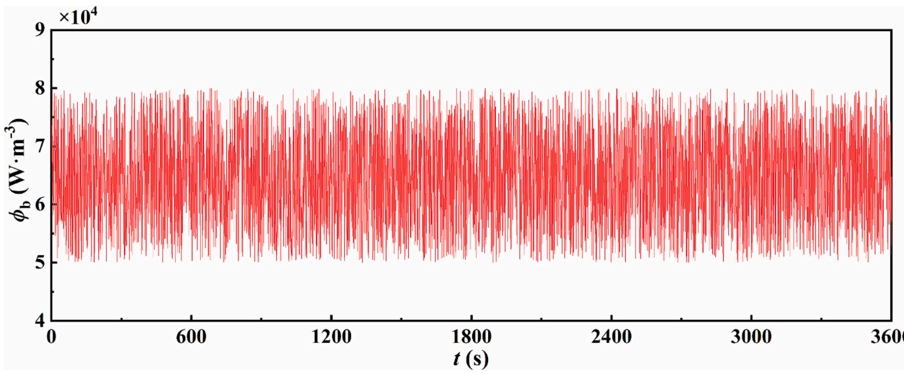


Fig. 10. Random heat generation rate with time for the batteries.

shown in Fig. 8 for comparing with the results by model. The results demonstrate that the ΔT results calculated by the developed model are in good consistency with those by CFD method. The deviations of the model results are less than 0.1 K. The CFD results show that ΔT during the whole discharge process is 0.91 K, which is lower than 1.0 K, indicating that the self-adaptive control strategy combined with the developed transient heat transfer model is efficient for regulating the temperature difference inside the battery pack below the preset value. Furthermore, the CFD results show that the average ΔT value is 0.48 K, and T is 331.2 K when the discharge process is finished. Moreover, CFD calculation is also carried out for BTMS J without flow pattern switch, with results shown in Fig. 8. For BTMS J, the average and maximum ΔT values are 0.99 K and 2.4 K, respectively, and the T_{max} is 332.0 K when the discharge process is finished. The average ΔT is reduced by 52% and T_{max} is decreased by 0.8 K after adopting the control strategy with ΔT_{lim} = 1.0 K.

Fig. 9 shows the temperature distributions of the battery pack at the end of the 5C discharge process for BTMS J and BTMS J-U-L with ΔT_{lim} = 1.0 K. It can be seen that after using the control strategy, the temperature uniformity of the battery pack has been significantly improved compared with that without flow pattern switch, and the maximum temperature is remarkably decreased. The results demonstrate that the developed heat transfer model is efficient to calculate the battery temperatures, and can be combined with the self-adaptive control strategy to switch the flow pattern of the system for reducing the temperature difference inside the battery pack. Moreover, the model needs much less time than the CFD method for calculating the battery temperatures, thus enable the real-time evaluation of the system performance and control of the temperature of the battery pack.

4.2. Varying operating conditions

In practical applications, the heat generation rate of the battery packs in an electric vehicle depends on the actual operating conditions. In this section, the condition of random heat generation rate for the batteries is considered to simulate actual operating conditions, and the heat transfer model combined with the self-adaptive control strategy is adopted to regulate the temperature difference of the battery pack. Considering the discharge time of the battery pack for 3600 s, the heat generation rate varies randomly between 5×10^4 W·m^{-3} and 8×10^4 W·m^{-3}, as shown in Fig. 10. This case is to simulate the varying operation conditions for electric vehicles.

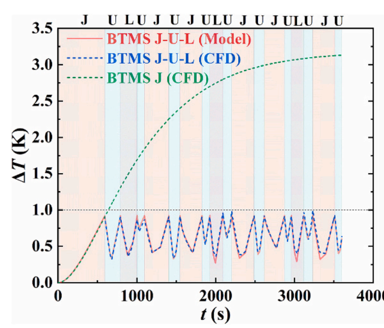


Fig. 11. Variation of ΔT with time at random heat generation rate (ΔT_{lim} = 1.0 K).

ΔT_{lim}, $\Delta\varepsilon$, δ, and Δt are set to 1.0 K, 0.01 K, 7%, and 1 s, respectively. Combined with the developed transient heat transfer model and control strategy, the regulation of ΔT in BTMS J-U-L is carried out. The moment at which the flow pattern switches (ΔT reaches 0.92 K) is predicted using the heat transfer model. The variation of ΔT with time is shown in Fig. 11. J-type flow is set initially for BTMS J-U-L. According to the results obtained by the transient heat transfer model, ΔT remains below 1.0 K during the whole operating process after switching the flow pattern 17 times. Then CFD calculation is carried out according to the control results, with ΔT results listed in Fig. 11 for comparing with the results by the model. Again, the results demonstrate that the ΔT results calculated by the developed model and CFD method are in good

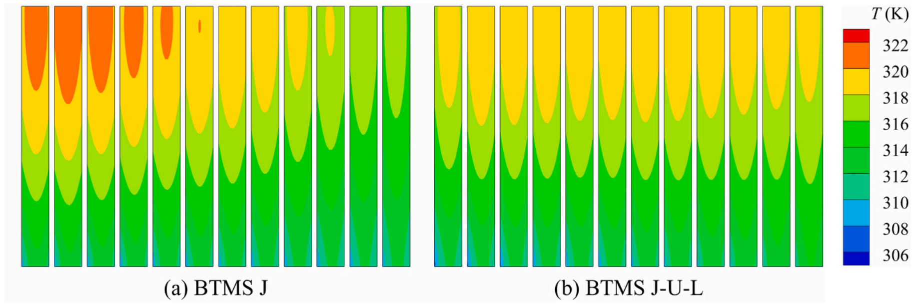


Fig. 12. Temperature contours of the battery pack under varying operating conditions.

consistency. The deviations of the model results are less than 0.1 K. The CFD results show that ΔT during the whole discharge process is 0.91 K, which is lower than 1.0 K, showing that the control strategy with the developed heat transfer model is efficient for controlling the temperature difference inside the battery pack below the preset value under the random varying operating conditions. Furthermore, the CFD results show that the average ΔT value is 0.59 K, and T_{max} is 319.7 K. The performance of BTMS J without flow pattern switching is also evaluated using CFD method under varying operating conditions, showing that the average and maximum ΔT values are 2.18 K and 2.4 K, respectively, and T_{max} is 321.2 K. The average ΔT is reduced by 73% and T_{max} is decreased by 1.5 K after adopting the control strategy with $\Delta T_{lim} = 1.0$ K.

Fig. 12 shows the temperature distributions of the battery pack when the discharge process is finished under varying operating conditions. It can be observed that with the implementation of the control strategy, the temperature variation of the battery pack in BTMS J-U-L is significantly improved compared with that in BTMS J, and the maximum temperature is also reduced remarkably. In summary, the developed simplified heat transfer model with the self-adaptive control strategy is efficient to regulate the temperature difference of the battery pack below the preset value under varying operating conditions.

5. Conclusion

In this work, an efficient transient heat transfer model is developed for operation optimization via flow pattern switch of battery thermal management systems (BTMSs) with different flow patterns based on the law of energy conservation, with its effectiveness verified by CFD method. Combination of the model with the self-adaptive control strategy allows the switch of the flow patterns in the BTMS with mixed flow patterns of J-type, U-type and L-type flow (BTMS J-U-L). The performance of the self-adaptive strategy is examined under 5C discharge process and varying operating conditions, respectively. The operation control results are validated using CFD method. The following conclusions are obtained.

1. The results of the developed transient heat transfer model are in good agreement with the CFD results. The relative deviations of the temperature difference inside the battery pack (ΔT) for the J-type, U-type and L-type systems are 4%, 2% and 7%, respectively, which demonstrates the efficiency of this model in real-time temperature distribution prediction of battery packs.
2. Under 5C discharge condition, temperature difference within 1.0 K is realized in BTMS J-U-L by using the transient heat transfer model and the self-adaptive control strategy for flow pattern switch. Compared to J-type BTMS with a single flow type, the T_{max} and average ΔT for BTMS J-U-L are reduced by 0.9 K and 72%, respectively.
3. Temperature difference is also maintained below 1.0 K under varying operation conditions in BTMS J-U-L. Compared to J-type BTMS with a single flow type, the T_{max} and average ΔT for BTMS J-U-L are reduced by 1.5 K and 73%, respectively.

In summary, the developed transient heat transfer model enables the efficient prediction of the real-time temperature and temperature difference inside the battery pack in different parallel air-cooled BTMSs, including the systems with different flow pattern, different numbers of battery cells, various types of batteries and various sizes. Combination of this model with the self-adaptive control strategy shows great potential for regulation of temperature difference inside battery packs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] X. Li, F. He, L. Ma, Thermal management of cylindrical batteries investigated using wind tunnel testing and computational fluid dynamics simulation, J. Power Sources 238 (2013) 395–402.
[2] F.P. Junaid, D. Rakshit, B.B. Saha, Converging passage: a strategy for uniform cell cooling in air-cooled battery thermal management system, Therm.Sci. Eng. Prog. 61 (2025) 103559.
[3] H. Cho, H. Lee, S. Ryu, Sequential multi-objective bayesian optimization of air-cooled battery thermal management system with spoiler integration, J. Energy Storage 114 (2025) 115586.
[4] C. Lee, Thermofluid topology optimization of liquid-based cooling plate for lithium-ion battery pack of EVs, Energy 323 (2025) 135712.
[5] L. Fu, Z. Zhang, L. Sheng, Z. Kuang, Z. Zhu, Pouch lithium-ion battery thermal management by using a new liquid-cooling plate with honeycomb-like fins, Case Stud. Therm. Eng. 69 (2025) 105945.
[6] G. Pulugundla, P. Dubey, A. Srouji, Time-accurate CFD analysis of liquid cold plates for efficient thermal performance of electric vehicle Li-Ion battery modules, SAE Tech (2019), https://doi.org/10.4271/2019-01-0500.
[7] W. Yang, C. Li, X. Li, H. Wang, J. Deng, T. Fu, Y. Luo, Y. Wang, K. Xue, G. Zhang, D. Zhou, Y. Du, X. Li, High flame retardant composite phase change materials with triphenyl phosphate for thermal safety system of power battery module, eTransportation 20 (2024) 100325.
[8] W. Yang, Y. Du, Q. Deng, S. Li, C. Li, J. Tian, P. Chen, T. Fu, Y. Luo, Y. Zhang, S. Zhu, X. Liu, Z. Rao, X. Li, High thermal conductivity composite phase change material with Zn_2^+ metal organic gel and expanded graphite for battery thermal management, Appl. Therm. Eng. 249 (2024) 123358.
[9] K. Chen, Q. Huang, Q. Li, S. Liang, X. Wu, Optimization strategy for battery thermal management system with phase change materials, aerogel and cold plates, Int. J. Heat Mass Transf. 221 (2024) 125070.

[10] J. Zhang, H. Zhang, J. Shi, S. Xu, C. Zhao, Performance study of a compact battery module integrated with thermal spreader and L-shaped heat pipes under high discharge rates, Appl. Therm. Eng. 270 (2025) 126258.
[11] H. Lu, M.M. Noor, K. Kadirgama, M.S. Beg, Startup, heat transfer, and flow characteristics of a ⊥−shaped oscillating heat pipe for application in a hybrid battery thermal management system, Appl. Therm. Eng. 269 (2025) 125984.
[12] H.T. Boateng, R.D. Jilte, A. Afzal, Numerical investigation of a cylindrical lithium-ion battery pack with integrated phase change material and coolant circulating channels, J. Energy Storage 73 (2023) 109441.
[13] R. Jilte, A. Afzal, S.A. Khan, M. Asif, E.P. Venkatesan, A. Munimathan, Cooling performance of a Li-Ion cylindrical battery pack with liquid circulating pipes embedded in phase change material, J. Energy Storage 87 (2024) 111335.
[14] Y. Liu, J. Zhang, Self-adapting J-type air-based battery thermal management system via model predictive control, Appl. Energy 263 (2020) 114640.
[15] P. Saechan, I. Dhuchakallaya, Numerical investigation of air cooling system for a densely packed battery to enhance the cooling performance through cell arrangement strategy, Int. J. Energy Res. 46 (2022) 20670–20684.
[16] K. Kirad, M. Chaudhari, Design of cell spacing in lithium-ion battery module for improvement in cooling performance of the battery thermal management system, J. Power Sources 481 (2021) 229016.
[17] X. Peng, X. Cui, X. Liao, A. Garg, A thermal investigation and optimization of an air-cooled lithium-ion battery pack, Energies 13 (2020) 2956.
[18] G. Zhao, X. Wang, M. Negnevitsky, H. Zhang, C. Li, Performance improvement of a novel trapezoid air-cooling battery thermal management system for electric vehicles, Sustainability 14 (2022) 4975.
[19] G. Singh, H. Wu, Effect of different inlet/outlet port configurations on the thermal management of prismatic Li-Ion batteries, J. Heat Transf.-Trans.ASME 144 (2022) 112901.
[20] F. Zhang, P. Liu, Y. He, S. Li, Cooling performance optimization of air cooling lithium-ion battery thermal management system based on multiple secondary outlets and baffle, J. Energy Storage 52 (2022) 104678.
[21] Y. Liu, J. Zhang, Design a J-type air-based battery thermal management system through surrogate-based optimization, Appl. Energy 252 (2019) 113426.
[22] W. Li, M. Xiao, X. Peng, A. Garg, L. Gao, A surrogate thermal modeling and parametric optimization of battery pack with air cooling for EVs, Appl. Therm. Eng. 147 (2019) 90–100.
[23] J. Zhang, X. Wu, K. Chen, D. Zhou, M. Song, Experimental and numerical studies on an efficient transient heat transfer model for air-cooled battery thermal management systems, J. Power Sources 490 (2021) 229539.
[24] J. Zhang, M. Song, X. Wu, Z. Zhang, K. Chen, Hot spot temperature optimization of turbulent heat convection systems: application to battery thermal management systems, Int. J. Heat Mass Transf. 242 (2025) 126779.
[25] C. Lyu, Y. Song, L. Wang, Y. Ge, R. Xiong, T. Lan, A new structure optimization method for forced air-cooling system based on the simplified multi-physics model, Appl. Therm. Eng. 198 (2021) 117455.
[26] F. He, H. Wang, L. Ma, Experimental demonstration of active thermal control of a battery module consisting of multiple Li-Ion cells, Int. J. Heat Mass Transf. 91 (2015) 630–639.
[27] X. Zhang, M. Li, Y. Zhang, F. Wang, K. Wu, Experimental and numerical investigation of thermal energy management with reciprocating cooling and heating systems for Li-Ion battery pack, J. Energy Eng. 144 (2018) 04018039.
[28] R. Mahamud, C. Park, Reciprocating air flow for Li-Ion battery thermal management to improve temperature uniformity, J. Power Sources 196 (2011) 5685–5696.
[29] W. Zhuang, Z. Liu, H. Su, G. Chen, An intelligent thermal management system for optimized lithium-ion battery pack, Appl. Therm. Eng. 189 (2021) 116767.
[30] S. Wang, K. Li, Y. Tian, J. Wang, Y. Wu, S. Ji, Improved thermal performance of a large laminated lithium-ion power battery by reciprocating air flow, Appl. Therm. Eng. 152 (2019) 445–454.
[31] K. Chen, Z. Zhang, B. Wu, M. Song, X. Wu, An air-cooled system with a control strategy for efficient battery thermal management, Appl. Therm. Eng. 236 (2024) 121578.
[32] H. Zhao, X. Fang, Temperature field simulation of lithium-ion battery pack for electric vehicle and its heat dissipation structure optimization, j, Hefei univ. Technol., Nat. Sci. 41 (3) (2018) 289–293.
[33] W. Wu, W. Wu, S. Wang, Thermal management optimization of a prismatic battery with shape-stabilized phase change material, Int. J. Heat Mass Transf. 121 (2018) 967–977.
[34] K. Chen, Y. Chen, Z. Li, F. Yuan, S. Wang, Design of the cell spacings of battery pack in parallel air-cooled battery thermal management system, Int. J. Heat Mass Transf. 127 (2018) 393–401.